UMKC Impact & Translation Series

Founding a Research-Focused Business

Helping you move your work beyond the university

Research at UMKC often begins with discovery, but many projects have the potential to create impact beyond the university. In some cases, that impact is achieved by forming a new venture.

This pathway is not required, and it is not the right fit for every innovation. However, when existing organizations are not well positioned to adopt or deploy a new idea, creating a business or nonprofit can become a practical way to move that work into real-world use.

This paper outlines how to think about that decision and what it takes to move from research to a formed entity.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: **when forming a business makes sense as a pathway to move your work beyond the university.**

The Evolution of Technology Transfer

Technology transfer in the United States has evolved significantly over the past few decades, laying the groundwork for universities to commercialize their research. A major turning point was the 1980 Bayh-Dole Act, which allowed universities to own patents from federally funded projects. This empowered academic institutions to create technology transfer offices to help researchers patent inventions and secure commercial pathways.

At UMKC, the Office of Technology Transfer (OTT) is responsible for evaluating, protecting, and managing intellectual property.

The Office of Commercialization (OC) works with faculty and graduate students to explore how research can move into real-world use. This includes helping assess pathways, supporting startup formation, and connecting researchers with external partners.

These offices work together, but serve distinct roles in the translation process. By engaging with UMKC's OTT and OC, researchers can lay a solid foundation for long-term business success.

Deciding to Spin Out: Key Questions

Not every innovation should become a company. In many cases, licensing to an existing organization or forming a partnership is a faster and lower-risk path to impact.

Forming a new venture becomes relevant when:

- No clear industry partner exists
- The technology requires significant development before adoption
- The opportunity is large enough to justify building new infrastructure around it

Before moving in this direction, it is important to evaluate both the opportunity and the level of commitment required. Consider the following:

- **Is there a clear market need?** - Even the most novel technology must fulfill a real customer or societal requirement. Programs like NSF I-Corps or engagement with UMKC's OC can help you identify genuine market opportunities.
- **Will development progress faster in academia or industry? -** Early-stage technologies may benefit from university labs and grants. However, transitioning to a startup can accelerate product development and commercialization efforts.
- **Are you prepared for the demands of entrepreneurship? -** Balancing academic responsibilities with a fast-paced startup may be challenging. High-growth companies often need a founder who can dedicate full-time attention to development and fundraising.

By weighing these questions upfront, you gain clarity on whether spinning out a venture aligns with your research timeline, risk tolerance, and ultimate professional ambitions. It's the first critical step in ensuring your journey is driven by realistic expectations and purposeful planning.

Publication versus Protection

Protecting intellectual property is not about limiting publication. It is about preserving the ability for your work to be developed, deployed, and used.

When researchers make a new discovery, the drive to publish and share results is both natural and central to academic progress. However, public disclosures—such as conference presentations, poster sessions, or journal publications—can inadvertently jeopardize patent rights if made before securing the necessary intellectual property (IP) protections.

Balancing the need for academic dissemination with the strategic timing of patent filings is crucial for preserving potential commercial value. To achieve this balance, consider:

- **Consult the Office of Technology Transfer (OTT) Early** - If you plan to present or publish findings that may be patentable, contact OTT well in advance. This ensures your team has enough time to file a provisional or full patent application before any public disclosure.
- **Understand Grace Periods vs. Global Rules -** While the United States allows a one-year grace period after certain disclosures, many international jurisdictions do not. If you are

aiming for global patent protection, securing your filing prior to any public release is essential.

- **Leverage Non-Disclosure Agreements (NDAs) -** In some cases, you can share details with collaborators or funding sources under an NDA. This keeps the information confidential and protects novelty if your patent has not yet been filed.
- **Plan for Conference and Journal Deadlines -** Conference abstracts and journal submissions can be considered public disclosures. Coordinate with OTT on filing strategies—such as provisional patent filings—that accommodate academic timelines while preserving patent rights.
- **Monitor Internal Presentations -** Even internal university seminars may be open to non-UMKC audiences, effectively making them public disclosures. Confirm audience details or establish confidentiality requirements in advance.

Being mindful of publication timelines and disclosure policies helps safeguard the novelty of your invention. With early coordination between you, your co-authors, and OTT, it is possible to share significant findings for academic advancement while still preserving the opportunity to pursue robust IP protections.

Navigating the Technology Transfer Process

Having established the viability of your idea and recognized the broader context of tech transfer, the next step involves officially safeguarding and licensing your intellectual property.

At UMKC, this process begins with disclosing your invention to the Office of Technology Transfer (OTT) and considering how it may eventually be licensed for commercial or nonprofit use:

- **Intellectual Property Rights** - If your invention was developed using UMKC resources, the university typically retains IP ownership. You will work with the OTT to file patents or other appropriate protections.
- **Licensing Agreements -** To launch a startup (or nonprofit) around UMKC-owned IP, you must license it from the university. Negotiation terms often include royalties, equity, or both, and a clear agreement can be pivotal for attracting investors or grantors.
- **Legal Considerations and Timeline** - IP filings and licensing negotiations take time. Consult legal counsel, maintain frequent communication with the OTT and OC, and plan for a period of back-and-forth before your venture officially forms.

By engaging with technology transfer professionals early, you protect your discoveries and lay legal groundwork that appeals to funders and partners. This proactive approach helps prevent delays and sets the stage for a successful spinout or licensing arrangement.

Once intellectual property is protected and a pathway has been selected, the focus shifts from protection to execution - building the structure, team, and strategy required to move the work forward.

Assembling the Founding Team

One of the most common misconceptions is that the researcher must also serve as the CEO. In many cases, the most effective role for a faculty inventor is as a technical founder or advisor, not an operational leader.

No venture thrives on technology alone. Successful startups and nonprofits require a blend of technical brilliance, business strategy, and day-to-day operational leadership. Building a cohesive founding team is often the difference between an innovation that stalls in the lab and one that scales in the market.

A successful venture—whether a for-profit or nonprofit—requires the right blend of skills and experience:

- **Technical Expertise -** Faculty members and graduate students intimately understand the science or technology behind the innovation.
- **Business Leadership -** Bringing on an experienced business strategist or entrepreneur can ensure savvy financial planning, market outreach, and operational setup.
- **Mentors and Advisors -** The UMKC Innovation Center, Missouri SBDC at UMKC, and Technology Venture Studio maintain extensive networks of professionals who can help guide first-time founders.

By selecting complementary team members and mentors from the outset, you create a collective knowledge base that can address both technological roadblocks and strategic hurdles. In turn, this balanced approach boosts your venture's credibility and momentum, especially with investors and early adopters.

Business Structures and Formation

Once you've gathered the right people, you'll need to choose a legal entity and formalize your operations. The goal at this stage is not to master corporate law, but to understand the implications of different structures and make an informed initial choice with appropriate legal guidance. The business structure you select influences governance, taxation, and even the type of financing you can pursue, making it a pivotal step in your commercialization journey.

Choosing the appropriate legal entity for your venture is foundational. Each structure carries unique implications for tax treatment, governance, and fundraising:

- **Common Business Structures:**
    - **LLC (Limited Liability Company)** - Offers flexible management arrangements and pass-through taxation, making it a popular choice for small teams.
    - **C-Corporation -** Standard for high-growth startups seeking venture capital, offering the ability to issue multiple classes of stock.
    - **S-Corporation -** Provides certain tax advantages but is subject to limitations on ownership and shareholder types.
    - **Nonprofit Organization (501(c)(3) or Similar) -** Ideal for ventures focused on public good rather than profit distribution. Qualifying for nonprofit status can open doors to

grants and charitable donations but restricts profit-sharing. This structure may also provide an avenue for large-scale social impact, particularly in fields like healthcare or community development.

- **State of Incorporation** - Many businesses incorporate in Delaware for its favorable corporate law, but local incorporation could be more convenient, particularly if you expect to remain regionally focused.
- **Corporate Governance -** Regardless of structure, establishing clear governance (board of directors, bylaws, etc.) helps ensure transparency, accountability, and investor or donor confidence.

Choosing the right entity requires balancing your growth ambitions, funding needs, and compliance obligations. It's worth seeking legal or accounting advice so you can begin on solid ground, ready for either investment or grant-seeking.

Planning and Strategy Development

A solid plan clarifies your mission, milestones, and resource needs. A well-structured plan or pitch deck is vital for communicating your venture's direction and viability to key stakeholders—co-founders, investors, partners, and even your internal team. Laying out coherent goals, timelines, and budget forecasts ensures everyone aligns on how best to move forward.

Whether you create a comprehensive business plan or a concise pitch deck, ensure you address:

- **Executive Summary** – One-page summary of the technology, target market or beneficiaries, and value proposition.
- **Market Analysis** - Define your total addressable market (TAM) or community need, identifying potential competition and detailing your unique advantage.
- **Technology and Product (or Program) Roadmap -** Map out how you will transition from research prototype to a fully deployable solution. Note any regulatory or compliance hurdles.
- **Financial Projections** - Include revenue models (for for-profits) or funding strategies (for nonprofits), along with budget forecasts and timelines. The Missouri SBDC can help refine these projections.
- **Risk Assessment and Mitigation -** Identify the largest technical, regulatory, or market uncertainties and detail strategies to minimize their impact.

By investing time in a thorough plan, you create a shared vision of success that resonates with funders and clarifies tasks within your team. Revisit this document as you gather market feedback, ensuring it evolves alongside new insights and growth opportunities.

Securing Funding

Funding is not the goal. It is a tool used to reach the next meaningful milestone.

Securing capital is critical to early-stage ventures. Securing capital fuels the transition from lab bench to market launch. Whether you opt for grants, equity investors, or philanthropic donations, diversifying your funding strategy typically yields the best outcomes.

A multi-pronged funding approach often proves most effective:

- **Angel Investors and Venture Capital -** Many startups progress from angel funding to larger VC rounds as they demonstrate market traction and scalability.
- **Federal Grants (SBIR/STTR) -** For tech-focused businesses, these non-dilutive funding sources can help move a lab prototype toward commercialization.
- **Foundation Grants and Donations (Nonprofits) -** Nonprofits can tap into philanthropic foundations, charitable donations, and government contracts aligned with their social mission.
- **Local and Regional Resources -** Kansas City boasts a supportive ecosystem, with incubators and pitch competitions regularly providing seed grants and business guidance. OC can help you discover which local programs fit your venture stage.

Selecting the right mix of funding channels can accelerate development, fuel pilot tests, and help you navigate the daunting "valley of death" between early R&D and consistent revenue. Matching your objectives with the right investor or grant source is critical to long-term sustainability.

Regulatory and Sector-Specific Guidance

Depending on your field, navigating regulatory frameworks can either be a minor concern or a central focus. Certain industries—like biotechnology or aerospace—carry strict standards that profoundly shape product development timelines.

Building a plan for compliance early prevents costly delays down the line:

- **Medical Devices and Biotech** - FDA regulations (e.g., 510(k), PMA) could require lengthy testing phases and specialized approvals.
- **Healthcare Software -** HIPAA or cybersecurity standards often apply if your technology handles protected health information.
- **Energy, Aerospace, and Defense -** Additional certifications or security clearances may be needed to operate within these sectors.

By factoring sector-specific regulations into your early planning, you avoid costly delays and enhance investor confidence. Engaging regulatory advisors or leveraging university support can smooth the path to compliance, ensuring your innovation meets all necessary standards.

Conflict of Interest and University Policies

When transitioning from academic researcher to entrepreneurial founder, it's vital to maintain transparency and uphold UMKC's guidelines. Carefully disclosing and managing conflicts protects both the university and your venture from legal or ethical complications.

To manage this transition, be aware of:

- **Disclosure Requirements** - UMKC requires faculty members to disclose external interests that may overlap with university-funded research or resources.
- **Use of University Resources -** Clarify if and how you can use UMKC labs, staff time, or equipment in your venture. Written agreements are often necessary for compliance.
- **Hiring Students and Postdocs -** Employing your mentees in a startup setting may raise ethical considerations. Ensure full transparency and follow university guidelines.

Guiding Question:

When does forming a new venture make sense as a pathway to impact, and what would it require to move your work in that direction?

By proactively addressing potential conflicts and adhering to UMKC policies, you safeguard your credibility and ensure that academic responsibilities and entrepreneurial ambitions remain harmoniously aligned.

UMKC-Specific Resources

UMKC boasts a robust ecosystem of offices and programs designed to guide researchers from idea to thriving venture. Faculty have access to additional supports that can significantly streamline business or nonprofit formation:

- **UMKC Innovation Center -** Offers training, workshops, and ongoing mentorship for idea validation, business model development, and strategic planning.
    - **Technology Venture Studio** - A specialized program aimed at accelerating technology-based ventures through mentorship, funding advice, and connections to industry partners.
    - **Missouri SBDC at UMKC -** Provides confidential, one-on-one counseling covering everything from financial projections to marketing plans.
    - **KCSourceLink -** Acts as a navigator to the Kansas City entrepreneurial ecosystem, connecting founders with capital sources, service providers, and community events. KCSourceLink
- **Pitch Competitions and Seed Funds**
  UMKC and local organizations frequently host events where you can gain visibility, funding, and feedback on your venture.
    - **Regnier Venture Creation Challenge –** An annual competition offering UMKC students, faculty, and alumni the chance to pitch new venture ideas. Winners often receive seed capital, mentorship, and broader visibility. This challenge can validate your concept and refine your pitch before external investor meetings.
    - **Entrepreneurial Innovation Grant (EIG)** - Provides financial support for early-stage ventures or proof-of-concept development. Ideal for covering initial prototype costs,

market research fees, or limited operational expenses that help you progress toward revenue or additional grants.

From formal IP management to nuanced mentoring and pitch competitions, these resources give you multiple avenues to refine your concept, rally partners, and secure funding. Tapping into them early often shortens the path to a successful business launch.

Common Misalignments

Early-stage ventures often struggle not because of weak technology, but because of misalignment between the opportunity and the approach. Common issues include:

- Pursuing a startup when licensing would be faster
- Building a company before validating real demand
- Assuming the inventor must lead the business
- Underestimating development timelines or regulatory complexity

Recognizing these early can prevent wasted time and increase the likelihood of meaningful impact.

Timeline and Key Milestones

Although every venture evolves at its own pace, establishing a general roadmap helps you manage progress, track goals, and pivot when necessary. This timeline is illustrative, not prescriptive. Progress depends on the nature of the technology, market readiness, and available resources.

Below is a high-level roadmap to keep your planning realistic and on track:

- **Months 0–6: Discovery & Ideation**
  - File an invention disclosure with OTT
  - Begin patent application if applicable
  - Conduct initial market or community need validation
- **Months 6–12: Formation & Licensing**
  - Select your legal structure (LLC, C-Corp, nonprofit, etc.)
  - Negotiate a license agreement with UMKC (if required)
  - Build the founding team
- **Months 12–18: Prototyping & Funding**
  - Develop a proof-of-concept or pilot program
  - Apply for SBIR/STTR grants or solicit angel investors (for for-profits)
  - Seek philanthropic or governmental grants (for nonprofits)
- **Months 18–24: Product/Program Refinement & Growth**
  - Expand beta testing or pilot projects
  - Pursue additional funding or strategic partnerships
  - Engage with the Technology Venture Studio, KCSourceLink events, or the Missouri SBDC to refine your go-to-market plan
- **Months 24+**
  - Launch product, service, or charitable programs to target audience
  - Scale team and operations

- Evolve funding strategy (VC rounds, donor campaigns, etc.)

Adhering to milestone-based scheduling keeps your team organized and investors informed, but remember this timeline is dynamic. Market feedback, regulatory shifts, or technical breakthroughs can all prompt adjustments.

Conclusion

Forming a business is one way to move research into real-world use. When aligned with the opportunity, it can create a focused path for development, funding, and deployment.

However, it is not the default path, and it is not required for impact.

By understanding how to navigate technology transfer, structure a business, secure funding, comply with regulations, and manage potential conflicts of interest, you set your enterprise on a path to sustainability. UMKC's ecosystem, including OTT, OC, the UMKC Innovation Center, Missouri SBDC, KCSourceLink, and dedicated grant programs. stands ready to assist at each pivotal stage. With careful planning, a committed team, and these institutional resources, your research can make a lasting impression well beyond the university walls.

Contacts & Resources

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